Supplementary Information

A. Supplementary derivations.

1. Full Expression for Equation (4) in the Letter.

We approximate ψ in Eq. (2) in the Letter up to the third-order in its Taylor expansion in x_0, around $x_0^{\,i}$, and perform straightforward integration in Eq. (2) in the Letter, to obtain an approximate expression

$$u(x,z) \cong \sum_i B_i Ai\left(\frac{k^{2/3}\left(\varphi''\left(x_0^{\,i}\right)+\frac{1}{z}\right)^2}{2^{2/3}\varphi'''\left(x_0^{\,i}\right)^{4/3}}\right). \tag{S1}$$

The coefficients B_i are given by

$$B_i = 2\pi e^{ik\left(\psi\left(x_0^i\right)+\frac{\left(\varphi''\left(x_0^{\,i}\right)+\frac{1}{z}\right)^3}{3\left(\varphi'''\left(x_0^{\,i}\right)\right)^2}\right)}\sqrt[3]{\frac{2}{\varphi'''\left(x_0^{\,i}\right)k}}\,. \tag{S2}$$

According to Eq. (5) in the Letter,

$$\left(\varphi''\left(x_0^{\,i}\right)+\frac{1}{z}\right) = 0 \quad . \tag{S3}$$

Accordingly, in this case

$$B_i = 2\pi e^{ik\left(\psi\left(x_0^i\right)\right)}\sqrt[3]{\frac{2}{\varphi'''\left(x_0^{\,i}\right)k}}\,. \tag{S4}$$

2. Derivation of equation (6) in the Letter.

In the Letter, we ensure that a global maximum of the intensity distribution exists for

all points on the curve $x = f(z)$ by requiring that there exists a smooth single valued relation $x_0(x)$ for all values of *x*. Consequently, $\partial x_0 / \partial x$ must be single valued for all values of *x*. We denote $z = f^{-1}(x) \triangleq g(x)$, drop the index *i* from $x_0^{\,i}$, and proceed by calculating the derivative $d\varphi'(x_0)/dx$ through the chain rule using $\partial x_0 / \partial x$ (which is only made possible since $\partial x_0 / \partial x$ is a single valued relation for all values of x). We obtain

$$\frac{d\varphi'(x_0)}{dx} = \frac{\partial \varphi'(x_0)}{\partial x_0}\frac{\partial x_0}{\partial x} = -\frac{1}{g}\frac{\partial x_0}{\partial x}, \tag{S5}$$

where in the right hand side of (S5) we used the relation $\frac{d\varphi'(x_0)}{dx_0} = -\frac{1}{z} = -\frac{1}{g(x)}$ from Eq. (5) in the Letter.

Next, we substitute $z = g(x)$ in Eq. (3) in the Letter, and directly differentiate Eq. (3), obtaining

$$\frac{d\varphi'(x_0)}{dx} = \frac{(1 - \partial x_0 / \partial x)\, g - \partial g / \partial x (x - x_0)}{g^2}. \tag{S6}$$

Equating the two expressions (S5) and (S6), we arrive at

$$x_0(x) = x - \frac{g}{\partial g / \partial x} = x - z\frac{\partial f(z)}{\partial z}, \tag{S7}$$

which is Eq. (6) in the Letter.

B. Derivation of the propagation distance at which diffraction-broadening onsets.

The beam may remain non-diffracting (more precisely, with no diffraction broadening) only as long as the experimental apparatus can produce the rays described by Eq. (6). Suppose that the phase mask has a finite size, *L,* and that the ray emanating from $x_0 = -L$ is tangent to the trajectory $x = f(z)$ at (x_L, z_L). Since $x = f(z)$ is convex, z_L is largest among all values of *z*, at which the other rays leaving the mask intersect the curve $x = f(z)$. For $z > z_L$, the ray tangent to *f(z)* intersects the plane z=0 below the edge of the mask ($x_0 < -L$). This ray cannot emanate from the phase mask: the mask is simply too short. Hence, there are no means to experimentally realize equation (6) in this case. Consequently, for a system with finite aperture *L*, normal diffraction-broadening will onset at any $z > z_L$. According to the above, z_L is obtained by substituting the trajectory $x = f(z)$, and the geometric position of the edge of the mask, $x_0 = -L$, in the ray equation (Eq. (6)), resulting in

$$\left.\frac{\partial f(z)}{\partial z}\right|_{z_L} = \frac{f(z_L) - (-L)}{z_L} \quad . \qquad \text{(S8)}$$

For any $z>z_L$, the beam no longer accelerates, and it strongly diffracts and broadens, as shown in Fig. 2d in the Letter. We note that the minimum width of the main lobe does not necessarily occur at z_L, and z_L is thus only an upper limit on the point at which diffraction broadening onsets. However, as we show below, z_L is practically close to the point of minimum width of the main intensity lobe.

We demonstrate the applicability and properties of Eq. (S8) with some examples. In Fig. 2b in the Letter, we presented the position of the peak intensity for a set of polynomial

beams that were pre-designed to intersect at z=150cm. Specifically, we produced optical fields following the set of trajectories

$$\begin{aligned} f_2(z) &= az^2 \\ f_3(z) &= bz^3 \\ f_4(z) &= cz^4 \\ f_5(z) &= dz^5 \end{aligned},$$

with the coefficients $a = 5\cdot 10^{-7}$, b=$3.33\cdot 10^{-10}$, c=$2.22\cdot 10^{-13}$,d=$1.48\cdot 10^{-16}$ (in millimetric units).

For a polynomial trajectory, $x = f(z) = rz^n$, with $n>1$ and when r is a non-vanishing coefficient, Eq. (S8) gives,

$$rnz_L^{\,n-1} = \frac{rz^n - (-L)}{z_L}$$

$$z_L = \sqrt[n]{L/\left[r(n-1)\right]}\,. \qquad \text{(S9)}$$

In the experimental apparatus presented in the Letter, the size of the SLM is L=12mm. With these parameters, Eq. (S9) predicts upper limits for the positions at which diffraction broadening onsets. These upper limits for the fields designed to follow the curves f_2 through f_5 are

$$\begin{aligned} z_L(f_2) &\cong 490cm \\ z_L(f_3) &\cong 262cm \\ z_L(f_4) &\cong 206cm \\ z_L(f_5) &\cong 182cm \end{aligned}$$

respectively.

We now test these predictions numerically. Figure 1S presents results of numerical simulations of the intensity distribution on the optical beams that were designed to

propagate along the functions f_2 through f_5. The intensity lobes of f_2 do not broaden while propagating from *z=0* to *z=300* cm (we have kept the intensity distribution presented below limited to 300cm to supplement comparison with the other intensity distributions). The intensity lobes of f_3, f_4, f_5 already start broadening at $z \cong 250, 200,$ and 180 cm respectively, in good agreement with the predictions above.

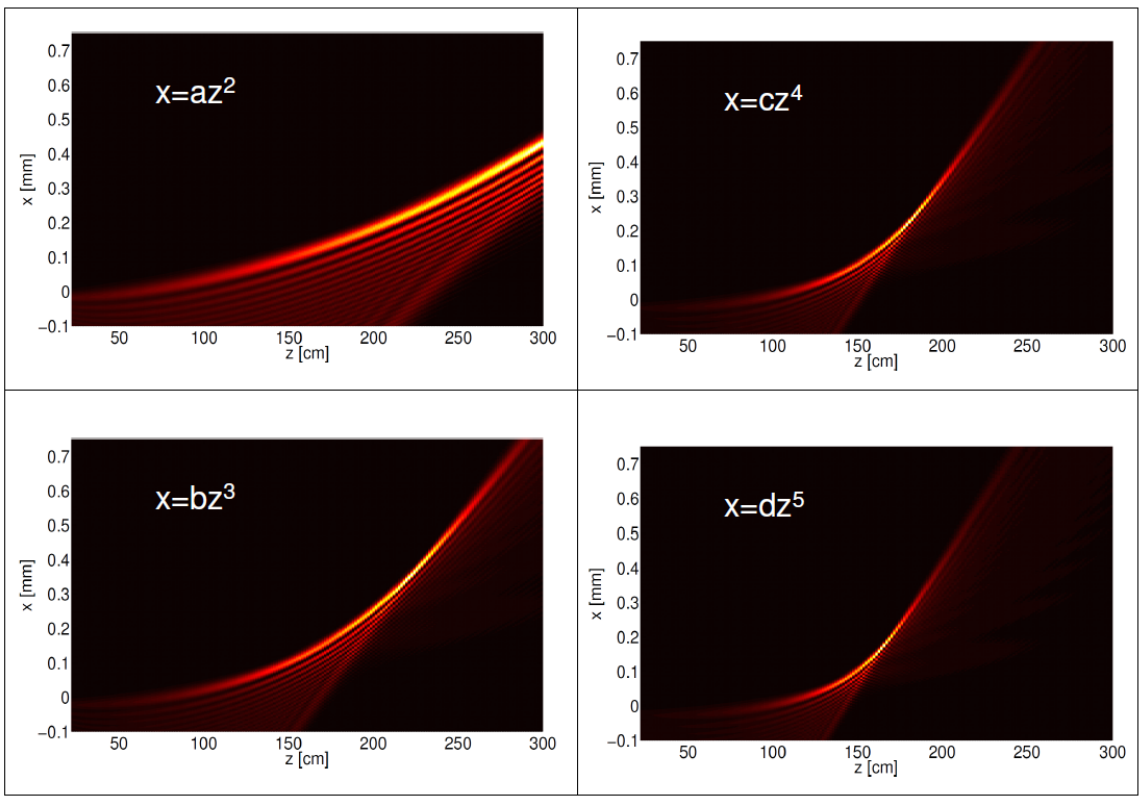

Figure 1S: Intensity distributions of optical beams emanating from a 12mm long phase mask positioned between (z,x)=(0,0) and (z,x)=(0,-12mm), and propagating along the trajectories $x = az^2, x = bz^3, x = cz^4, x = dz^5$ **.**

To obtain a more quantitative measure of the diffraction broadening, we present, in Fig. 2S the full-width-half-max (FWHM) of the peak-intensity lobe for the intensity distributions presented in Fig. 1S, as a function of the propagation distance, *z*. This figure

is similar and supplements Fig. 2d in the Letter. Evidently, for each intensity distribution, the main lobes narrow down at some propagation distance, and then begin to broaden. The beam following f_2 does not broaden at all for z<300cm , while for the other examples, the positions at which broadening onsets are $z_L = 170$, 189, and 232 cm for f_3, f_4, f_5 respectively, in agreement with the upper limit set by Eq. (5). The beams presented in the Letter and in Figs. 1S, 2S become as narrow as 100μm FWHM, but are non-broadening up to distances of 170-400 cm, that is, up to propagation distances ranging between 15 and 30 Rayleigh ranges of a Gaussian beam with an equivalent FWHM.

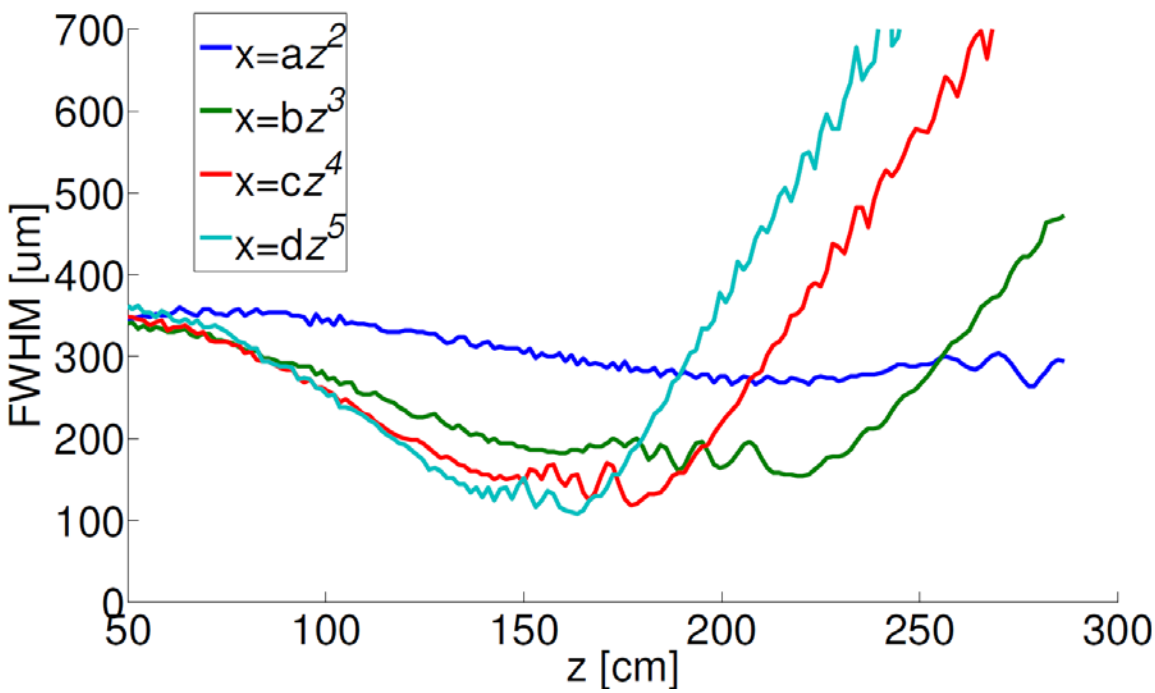

Figure S2: FWHM of the intensity distributions of Fig. 1S, vs. the distance of propagation, showing no diffraction broadening for > 30 Rayleigh ranges for $x = dz^5$.

C. Light beams bending exponentially in space.

As explained in details above, the finite size of the input phase mask limits the distance the beam is propagating until its intensity features stop bending, and diffraction-broadening onsets. This effect is particularly important for strongly bending trajectories such as $x = b\left[e^{kz} - 1\right]$ (presented in Fig. 2a in the Letter), where for small input aperture sizes - the strong exponential bending occurs only for a short propagation range.

Figure S3 presents a data set supplementing the exponentially curving intensity patterns presented in Fig. 2a in the Letter. This data set presents numerically obtained intensity patterns for exponentially-bending beams generated by phase masks with apertures sizes 20mm, 40mm, 60mm, and 80mm. The green curves are the pre-designed function $x = b\left[1 - e^{-kz}\right]$, used to generate the phase masks at z=0. This curve is redrawn for each of the intensity patterns, highlighting the range up to which the intensity features of the beam follow this pre-designed trajectory. The range over which the diffraction patterns follow the exponential trajectory increases with the size of the aperture. For instance, for an aperture size of 20mm, the exponentially bending intensity pattern stops bending at z~180cm, x~3mm, whereas for an 80mm aperture, it follows the pre-designed exponential function up to z~225cm, x~12mm. This example highlights the effect of aperture size on the resulting diffraction pattern, and further demonstrates our ability to generate exponentially bending light-caustics.

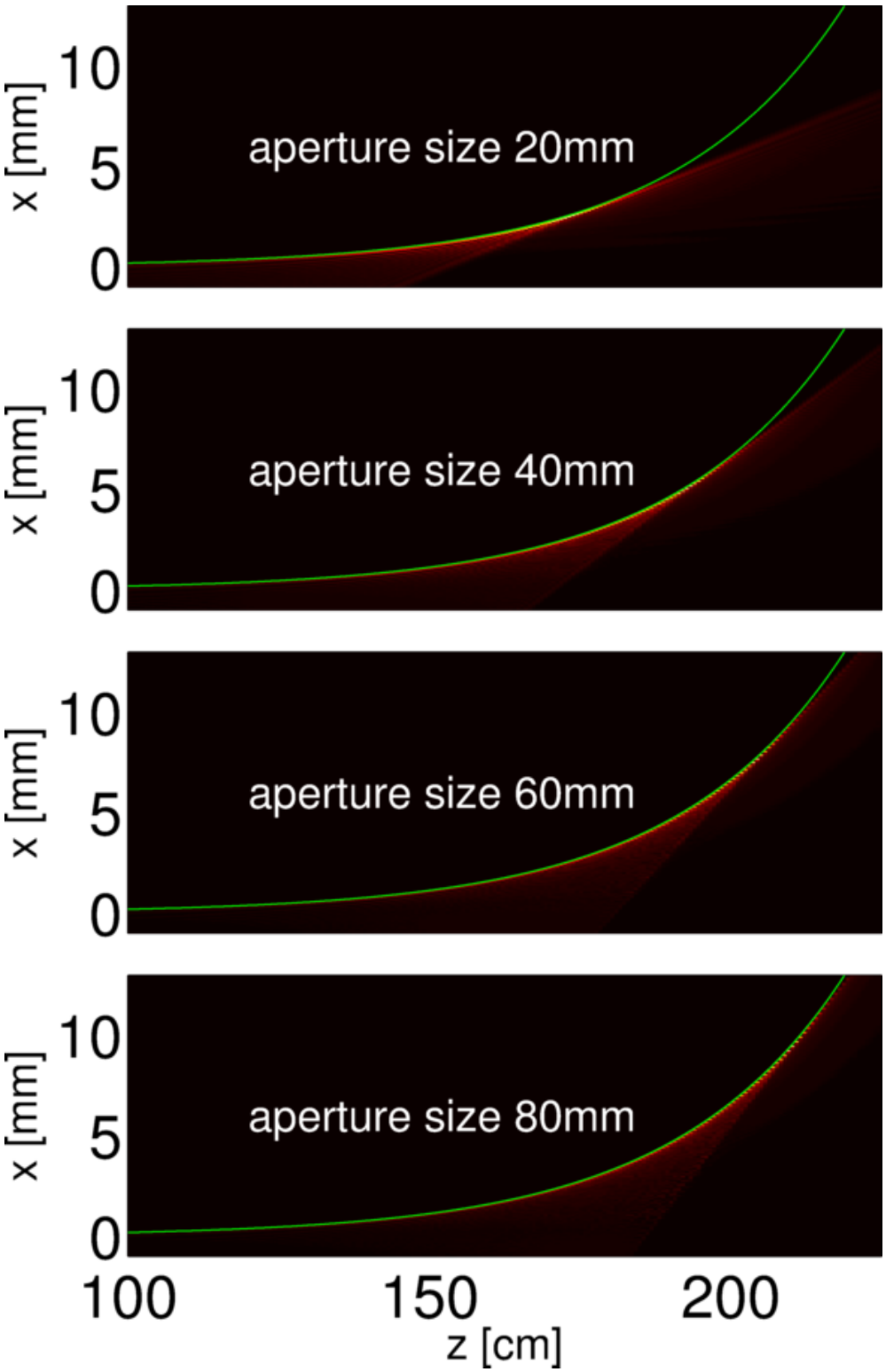

Figure S3: Exponentially bending intensity distributions, following the pre-designed trajectory $x = b\left[1 - e^{-kz}\right]$, for different sizes of the phase-mask apertures. The range up to which the diffraction patterns follow the exponential trajectory increases with the size of the aperture.